

Exploring the dynamic symbiosis of urban mobility and 5G networks[☆]

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ABSTRACT

The interdependence between urban mobility and 5G networks can bring several advantages for both domains. By exploring this dynamic symbiosis, we can uncover opportunities to enhance the performance, efficiency, and safety of urban transportation systems while leveraging the capabilities of 5G networks to provide strong connectivity, high data rate, and low-latency communications. This work explores their relationship and shows that we can use the urban mobility data of vehicles on the roads to predict the mobile communication network usage, and the opposite, the network data to predict the urban mobility. We analyze the correlation between urban mobility and the mobile communication network usage, finding strong correlations between the number of vehicles in each road direction, measured by the radars, and the usage of 5G base stations nearby. We then use the information from the radars data to predict handovers between different 5G gNBs and the network traffic, and vice versa, using techniques like LightGBM. We generate a mobility metric using Principal Component Analysis (PCA), and we infer the mobility data from 5G network data and vice versa, creating areas of interest by grouping nearby 5G stations and radars. We observe that, in most cases, we can achieve good results in the inference and prediction using LightGBM. This is extremely relevant to adapting the network resources in dynamic 5G slices while also predicting urban load and adapting the traffic management on the roads.

1. Introduction

We live in an interconnected world, and therefore, we can find an interdependence between the mobility of the users and their network services and connectivity. The ubiquity of smartphones, smartwatches, tablets, laptops, and other smart devices has revolutionized how we engage with information, services, and with each other. These devices rely on networks and telecommunication technologies to exchange data and communicate. The information about mobility and the one about 5G communication networks can benefit from each other, and this interdependence can lead to further advancements in both fields.

Monitoring and characterizing the movement of people within urban areas is an essential yet challenging task. Cities are implementing

strategies such as improving public transportation by enhancing planning, real-time traffic management, improved emergency response, and personalized mobility recommendations to address transportation challenges. Nevertheless, for this to be possible, it is necessary to measure people's mobility, which requires tracking people and conducting massive surveillance to achieve that. In this context, several devices and sensors can be used, such as video cameras [1], GPS sensors [2], traffic counters [3], RADARs and LIDARs, and even telecommunication data [4,5]. However, this brings significant issues related to privacy, besides the complexity and high costs of such solutions. Choosing a less intrusive way to track people's mobility is still preferable, since it is cheaper and can be used without privacy issues. Nevertheless, it can

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be less accurate. One possible solution is to use anonymous telecommunications data, through 5G cellular communication, to monitor the users' location concerning to which 5G cell the user is connected.

Fifth Generation (5G) technology can support billions of devices connected to the Internet without compromising the user's experience. With the incorporation of the 5G network function (Network Data Analytics Function (NWDAF)) in 5G networks, we can use data analytics and machine learning to improve the NWDAF, namely by estimating the users' mobility in the context of 5G networks. For instance, we can optimize the resource allocation, dynamic 5G network slices, improve the Quality-of-Service (QoS) and user experience, optimize the energy consumption, and target capacity expansion by predicting the number of users per base station and their needs. Therefore, with the users' mobility information, we can predict the requirements in terms of the network in its closest 5G cell.

As we mentioned in our previous work [6], there is a shortage of public real-world 5G datasets from 5G commercial networks. This results in a lack of studies related to 5G, especially works that relate 5G with the urban mobility. Similarly, the mobility data of vehicles in the roads is also very scarce in the cities, due to the strong sensing requirements. However, since mobility and network data can be interrelated, it may be possible to infer one from the other in locations where only one type of data is available. Using both, we may develop models to be used in different types of areas and cities.

Using mobile network data, such as 5G, to assess people's mobility offers several opportunities and benefits. Nowadays, most cities have 5G networks, and the number of devices supporting this technology is also increasing as people replace their old devices with new ones. This means that, using 5G network data to measure mobility is becoming increasingly more reliable. Furthermore, using 5G data to assess mobility enables the creation of synergies with other smart cities and provides homogeneity in urban mobility studies. Given the 5G network characteristics, such as low latency, 5G data can be obtained in real-time, allowing real-time analysis and predictions.

In this work, we explore the dual and dynamic relationship between urban mobility and the usage of 5G networks while preserving people's privacy. We start by analyzing the correlation between urban mobility and the network; namely, we correlate the number of vehicles counted in each direction from each vehicular radar on the road with the data from nearby 5G network base stations, such as data about the connected users and the handovers. We observe strong correlations between the data from the radars and the base stations. Then, we show that it is possible to take advantage of the relationship between the vehicles' mobility on the road and the usage of the 5G communications network to predict each other. Specifically, we use the information from the radars to predict the intra- and inter-gNB handovers, and the network traffic load in the nearby areas, and we use the information from the network traffic load to predict the number of vehicles in the nearby roads. We achieve a good accuracy in the prediction by using techniques like Light Gradient-Boosting Machine (LightGBM).

To infer the mobility data from the 5G network data and vice-versa, we start by creating areas of interest by aggregating nearby 5G stations, considering the city morphology and the correlation between the 5G base stations and the information from the radars. To address the inference of both the mobility and 5G network usage, we grouped RADARs and 5G network stations into zones and created a mobility metric, a global mobility metric, and a global 5G network metric. In a separate training, the selection of the best model is very biased by the available data. In a joint training, we visualize the expected patterns from the generated values from both 5G network usage and mobility, and we show the benefits of studying and relating mobility and 5G network data. We also show that LightGBM produces good results in most cases, outperforming the other methods. With this work, we present a comprehensive framework for the interaction between mobility and 5G data, including a multi-metric imputation approach to predict metrics when sensing devices are unavailable. To the best of

our knowledge, this is the first 5G network and mobility data approach, with real data from both the 5G mobile network operator and from the city infrastructure.

The main contributions of this work are the following:

- Data exploration pipeline for mobility and 5G network metrics;
- The dual and simultaneous exploration and analysis of mobility data and 5G network data and its correlation;
- Research of an approach to predict the 5G network using mobility data and vice versa;
- Research of 5G network load and mobility metrics;
- Research of an approach to infer the 5G network data using mobility data and vice versa when the information is not available on those areas;
- Extensive experiments with several machine learning approaches.

The remainder of this article is organized as follows. Section 2 presents the related work. Section 3 contains the adopted methodology. Section 4 presents the correlations between mobility and the network. Section 5 explores predicting mobility using the network usage and vice-versa, and Section 6 explores inferring mobility using the network usage and vice-versa. Finally, Section 7 contains the conclusions and ideas for future work.

2. Related work

This section presents related works on analyzing, characterizing, and studying urban mobility using telecommunications data.

There are several types of telecommunication data, such as data from Bluetooth sensors, Wi-Fi, or cellular stations. Nowadays, we can find Bluetooth or Wi-Fi scanners installed in fixed locations [4,5,7,8] or mobile locations [9,10], such as public transportation (trains, buses, and airplanes). Both Bluetooth and Wi-Fi detect nearby devices similarly. For instance, we can use Bluetooth discovery to scan the devices surrounding us with Bluetooth activated. Regarding the Wi-Fi, even if the mobile devices are not connected to any Wi-Fi Access Point (AP), they still broadcast Wi-Fi probe requests, enabling Wi-Fi sniffing. Therefore, Bluetooth and Wi-Fi allow a non-invasive study of people's mobility. Furthermore, these types of devices are easy to deploy and cost-effective.

Using fixed scanners in different locations, Lee et al. [4] used Bluetooth sensors to gather prevailing traffic conditions by capturing the MAC addresses of Bluetooth in mobile devices and car navigation systems. In [5], the authors analyzed crowd mobility, namely their arrival and departure behaviors. Furthermore, they also analyzed smartphone manufacturers and the transition patterns between locations. Salas et al. [7] proposed three methodologies to study urban mobility in near real-time using Wi-Fi data and GIS information. The first one is the classification of devices by frequency of occurrence. The second one is the estimation of transportation modes. The most frequent transportation mode identified is pedestrians, followed by people moving quickly using bicycles and scooters. The third one is estimating the most common travel routes using association rules. Traunmueller et al. [8] analyzed the trajectories of citizens using data from public Wi-Fi access points spread across the city. For this study, they removed data from stationary devices or if those devices did not present any movement for long periods. To validate the study, they aggregated the data and compared it to manually collected pedestrian counts. Nevertheless, they were not able to differentiate between pedestrians and non-pedestrians, meaning that there can be some inconsistencies. Then, they detected and analyzed trajectories, including their frequency.

Beyond static stations, we can also have mobile stations by installing the sensors in moving vehicles. Filgueiras et al. [9] used Bluetooth sensors deployed in vehicles to detect traffic conditions. With this, the authors created Origin-Destination matrices and did some studies regarding the number of devices detected throughout the day, the percentage of devices detected per trip, the activity captured by Bluetooth

devices, and estimated travel times. The authors of [10] inferred Origin-Destination matrices of public transportation passengers by tracking the passengers with an activated Wi-Fi interface. The authors ensured the passengers' privacy by avoiding the storage of device-specific information. The authors developed a validation algorithm to confirm whether the device was on the bus.

Certain works rely on data obtained from cellular networks. Calabrese et al. [11] used mobile phone trace data to study individual mobility patterns. The dataset contained anonymous location estimations from devices generated each time a device connects to the cellular network, such as receiving or making a call, receiving or sending a message, and accessing the internet. With this dataset, the authors were able to infer trips if there is enough data from each user. They also estimated the home location of those users by dividing the map in a grid of 500 m by 500 m, and analyzing the frequency of each user in each zone during the night. Using vehicle safety inspection data, the authors also estimated the annual kilometers traveled for every private passenger vehicle registered in the area. With this, the authors created two mobility metrics: an individual mobility metric (average daily total trip length of users) and a vehicular mobility metric (average daily kilometers traveled per vehicle). These metrics allowed the users to validate the study by comparing individual mobility with the odometer readings from annual safety inspections.

Sun et al. [12] used Fourth Generation (4G) cellular network data to characterize user mobility. They analyzed the occurrence of users connected to the 4G stations. They also measured the user mobility by computing the radius of circles. Furthermore, they studied user-dominant locations. The authors also concluded that 4G data had finer-grained mobility and location tracking than 3G data.

Vidović et al. [13] proposed a methodology for detecting public transportation modes using a dataset from anonymized telecom data and statistical modeling. The dataset contained user migrations, including the trip's origin, destination, and other relevant information. The public transportation mode was identified by analyzing possible stops and lines. This methodology was validated in a real scenario.

We can also find some works in which the authors take advantage of machine learning and the relationship between the network usage and mobility data to study, characterize, and improve them. Gebrie et al. [14] studied the performance of machine learning algorithms to predict mobility in cellular networks. The authors started by generating a realistic synthetic dataset using the Self-similar Least Action Walk (SLAW) mobility model [15], which mimics the actual behavior of cellular network users. They evaluated four algorithms: Deep Neural Network (DNN), eXtreme Gradient Boosting (XGBoost), Semi-Markov, and Support Vector Machines (SVMs). The authors concluded that XGBoost outperformed all the other algorithms, achieving a higher accuracy and allowing higher energy saving in Self-Organizing Networks (SONs).

Mobility prediction can also be helpful to improve the communication network knowledge. Jeong et al. [16] introduced two use cases for 5G networks using datasets containing 4G data. The first use case is to use machine learning to assist adaptive paging, the process of notifying the user equipment in idle mode by sending wake-up messages. The authors replaced the subtask of paging a list of previously visited nodes to page only the ones based on predicted future trajectories. They tested some models, with Recurrent Neural Networks (RNNs) being the most accurate, with a 75% gain. The second one is the User Plane Function (UPF) re-selection. Suppose the network knows beforehand that a user's equipment will move to a target site. In that case, some aspects of the relocation process can be completed prior to the handover, shortening the process. For this case, they use a multi-task Long Short-Term Memory (LSTM) to perform the prediction. Meneghello et al. [17] estimated the potential of predicting the next base station before the handovers by assessing mobility trajectories in 5G vehicular networks. The authors simulated mobility trajectories using the Simulation of Urban MObility (SUMO) simulator [18], and data from mobile millimeter-wave devices.

Regarding the prediction algorithm, they used markov Chains with RNNs and Convolutional Neural Networks (CNNs).

In our previous research [6], we studied and analyzed a dataset containing real 5G commercial data. We proposed the creation of a metric to measure the traffic load and using forecasting techniques based on LightGBM to help understand the trends and manage 5G networks. This work demonstrated the practical application of machine learning models in the context of NWDAF, which supports dynamic network slicing for 5G network expansion and deployments.

Even though several works take advantage of network data to predict mobility, the characteristics of 5G networks make them more suitable for this task. With 5G networks, we can collect data from a larger and more diverse range of devices, such as smartphones, vehicles, and wearables. This allows us to have a more comprehensive view of the mobility model. Besides, since 5G networks present a lower latency when compared to other networks, we can take advantage of real-time algorithms. Furthermore, 5G networks can also use mobility data to manage their resources, such as improving network slicing.

Moreover, this paper addresses not only how to predict mobility using the 5G data, but also how to predict the 5G data and its requirements with mobility. Finally, we provide a complete framework for the interaction between mobility and 5G data and the prediction of one of the metrics when no sensing devices are available, resulting in a multi-metric imputation approach.

It is important to note that using mobile communication networks to measure people's mobility has some limitations. For instance, not everyone carries their mobile devices with them all the time, and some people may have more than one device. Besides, people can have Wi-Fi and Bluetooth turned off. Additionally, we cannot monitor all network operators, as there are several of them. Moreover, only a portion of users have access to 5G technology, so some devices still rely on Third Generation (3G) or 4G for communication. Moreover, it is also difficult to have mobility data through mobility sensors in all locations in an area. Therefore, our approach is to understand the correlation between the 5G mobile network and mobility data, and research the inference and prediction that can be made using both 5G mobile network or mobility data to perform a complete map and prediction of both mobile network and mobility with incomplete data availability.

3. Methodology

This section presents the methodology used to interrelate both mobility and mobile network data. It starts by describing and illustrating the datasets used, then discusses the use cases drawn, and finally discusses model selection and evaluation.

3.1. Available datasets

This work considers two types of datasets: one contains mobile network data from a commercial 5G network, and the other contains mobility data from RADARs installed in a city. For both datasets, we have data from the 15th of September 2023, until the 30th of December 2023. We only have access to historical data, meaning that no real-time experiments are possible, but the data can be used to real-time predictions.

The commercial 5G dataset is from the same set of thirteen stations used in our previous work [6]. These stations were named from A to M. Therefore, we used the same processing that was described in that work. This dataset contained hourly data from several 5G metrics. The provided data from this dataset is aggregated, meaning that the dataset is completely anonymized, which does not allow the identification or tracking of users. We pre-selected fifteen 5G network metrics from this dataset, such as the connected users (average and maximum values), the handovers (inter- and intra-gNB), the average resource block used, the average user throughput, the user data volume, the user transmission time, and the cell transmission time. By applying

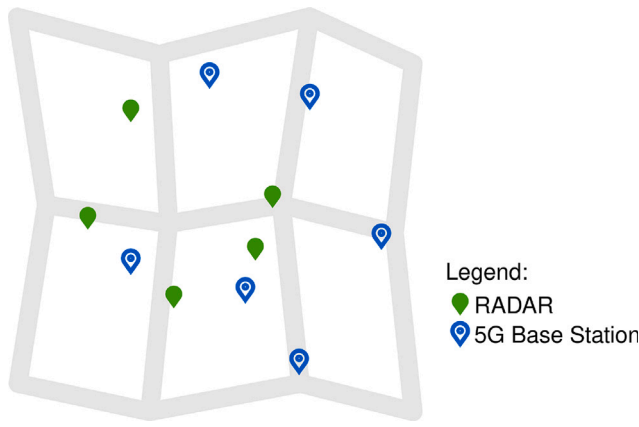


Fig. 1. Schematics of the location of both datasets, focusing on the city center.

Table 1

Use cases.

Use case	Description
Use the relationship between urban mobility and 5G networks to make predictions about each other	
Use case 1	Use RADAR data to predict intra and inter-gNB handovers
Use case 2	Use RADAR data to predict network traffic load
Use case 3	Use network traffic load data to predict RADAR data
Use the relationship between urban mobility and 5G networks to infer about each other	
Use case 4	Infer network traffic load from mobility data
Use case 5	Infer mobility data from network traffic load

Principal Component Analysis (PCA) to this dataset, we created a metric to measure the network traffic load.

Considering the mobility data, we received raw data from five RADARs in the city of Aveiro, Portugal, that included a timestamp, entity ID, heading, radar vehicle ID, vehicle class, speed, and acceleration. These RADARs were named from V to Z. Using the entity ID, we determined the location of the RADAR. By analyzing the heading, we were able to differentiate between road directions. To make the data more manageable and to compare it with the data from the 5G network, we aggregated the information into 1-hour intervals. Furthermore, we decided to create a metric to give the number of vehicles per hour and per road direction.

Even though mobility data is aggregated in hourly intervals, we can still conduct a macro analysis of mobility. Geographically, the sensors from the radar dataset are located near some of the stations from the mobile network dataset. Fig. 1 contains schematics of the location of the stations and the sensors from both datasets, focusing specifically on the city center. As we can visualize, we can observe that the figure contains all the five RADARs and only 6 of the 13 base stations.

3.2. Use cases

With this work, we aim to explore the relationship between urban mobility and 5G network usage. Therefore, we selected five use cases to evaluate the ability to predict network metrics using mobility data and mobility metrics using network data. These use cases are described in Table 1.

The first three use cases aim to make predictions about each other, such as using one to impute missing data or to forecast the other. For instance, considering that there is an energy failure that affects all the RADARs, if we know the relationship between network usage and mobility, we can predict the missing data from the RADARs based on the network usage's observed patterns.

The last two use cases aim to infer one from the other. Considering that we want to install a new network station in a new area, it might be useful to predict the usage of that new station before the installation to understand if there is an actual need and which resources will be needed. Considering the existence of RADARs in that new area, we can infer about each other by knowing the relationship between the RADARs and the network usage.

3.3. Model selection and evaluation

Several models can be used in regression tasks, starting from simple linear models to complex deep learning algorithms. We evaluated 7 different models: Linear Regression (LR), Ridge Regression, ElasticNet, Decision Tree (DT), Random Forest (RF) [19], XGBoost [20], and LightGBM [21]. LR, Ridge regression, and ElasticNet are linear models [19]. LR is based on the ordinary least squares, Ridge regression is similar but applies L2 regularization, and ElasticNet applies L1 and L2 regularization. DT and RF are tree-based models in which RF is an ensemble method [19]. XGBoost is built on the Gradient Boosting framework and implements machine learning algorithms, providing parallel tree boosting, making it an efficient option for a variety of applications. LightGBM [21] is a highly efficient and distributed framework for gradient boosting that utilizes tree-based learning algorithms such as GBDT, DART, and GOSS. This framework is designed to work with large datasets using low memory. LightGBM proved very effective during the fifth edition of the forecasting accuracy competition [22] even when competing with more complex models, such as deep learning models.

For each model, we consider different parameters, except for LR. For Ridge regression, we change the value of alpha and the solver. For the DT, we experiment with different values for the maximum number of features, maximum depth, minimum samples to split, and minimum samples per leaf. For the RF, we change the number of estimators, maximum number of features, maximum depth, and the criterion. For the XGBoost, we consider the number of estimators, maximum depth, learning rate, subsample, and the subsample ratio of columns when constructing each tree. Finally, for LightGBM, we consider the same parameters and values as in XGBoost. Furthermore, we add the variations of the boosting type, number of leaves, and lambda values.

Considering that the data available corresponds only to three months, there is a limited volume of data to feed the models. Deep learning methods imply a higher volume of data to be available to optimize all parameters in a high number of layers [23,24]. Therefore, in the scope of this work (a proof of concept), traditional machine learning models are used. This approach will allow us to infer and show the viability and applicability of the concept.

Considering the regression problem, and to keep the consistency with our previous work [6], we select two metrics to evaluate the performance of the models: the Root Mean Squared Error (RMSE) and the R^2 -Score. Note that LR, Ridge, and ElasticNet are deterministic models. Additionally, we calculate the Average RMSE and the Average R^2 -Score to compute the average performance of the models applied to different stations and sensors.

4. Correlation between mobility and network data

We start by exploring the relationship between each 5G network station and the radars. PCA 0 and PCA 1 are the first and second PCA components that result from the application of PCA to the 5G network dataset. Fig. 2 contains part of the correlation matrix for station B.

We zoom on the correlation matrix to highlight the most informative correlations. We decided to keep only the correlation of the information from the radars, that is, the number of vehicles measured in each direction, with the communication network features. As we can observe, radar Y (both lanes) and radar Z (lane 1) are the ones that achieve a more significant correlation with station B. This can be observed for almost every field except the handovers intra-gNB,

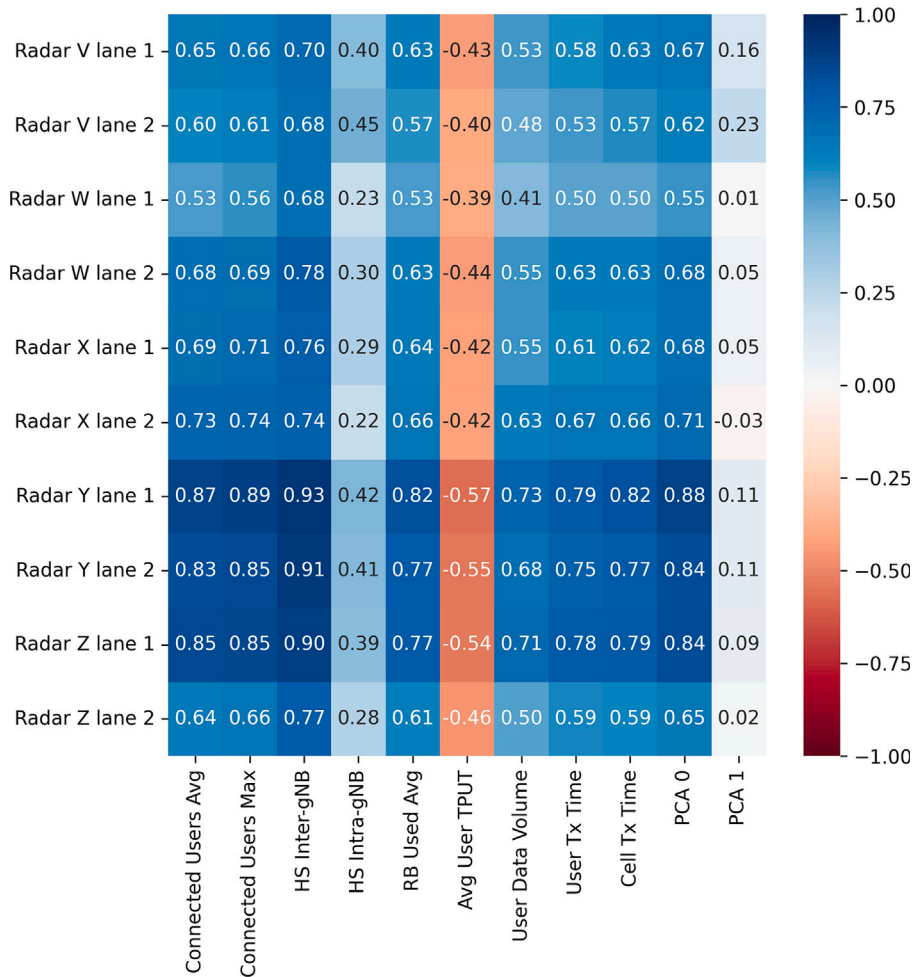


Fig. 2. Correlation matrix between base station B and the RADARs (zoom on rows and columns of interest).

the average user throughput, and the second component of PCA (PCA 1). Furthermore, the first component of PCA (PCA 0) presents similar values to almost all the other 5G metrics. This supports our choice of using this component to represent the network traffic load. The strong correlation between the first component of PCA with the number of vehicles on some of the radars indicates that there is a strong potential in exploring this relationship to improve urban mobility and network resource allocation.

In a 5G base station, a gNodeB (or gNB) contains several cells. As we can observe in Fig. 3, the inter-gNB handovers are performed between cells of different 5G stations. As depicted in Fig. 4, the intra-gNB handovers are performed between cells of the same station. When a user moves from one location to another, they may need to connect to a different station. As expected, the inter-gNB handovers present a stronger relationship with the data from RADARs than the intra-gNB handovers since they capture better the user mobility.

The correlation between mobility data and network data suggests a meaningful relationship between the two: as one increases, the other generally increases as well. However, despite the strong correlation, the correlation coefficient is not exactly 1, indicating that a purely linear relationship does not exist. Therefore, a simple linear function cannot precisely predict one variable from the other. Lower correlation values in certain samples are expected, as the relationship between mobility and network data is not perfect. Several factors contribute to this variation, including:

- stationary pedestrians, whose mobility may not align with network activity,

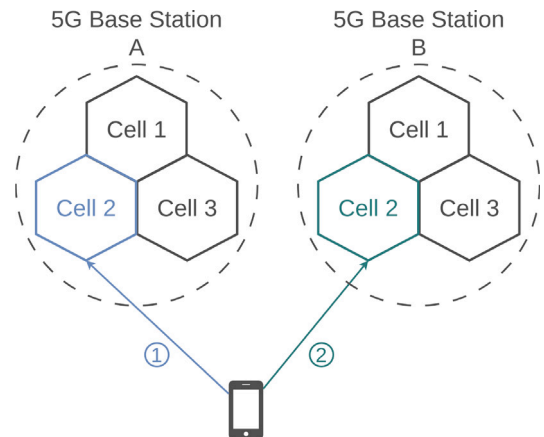


Fig. 3. 5G inter-gNodeB handovers.

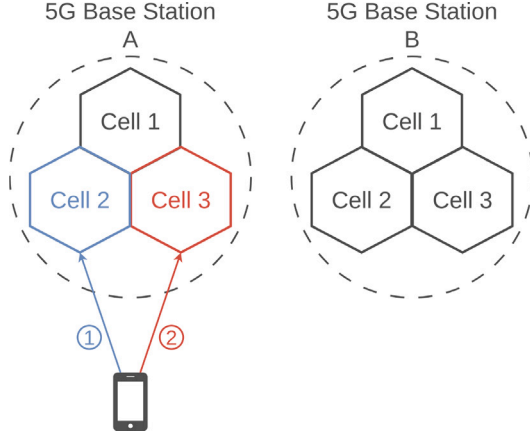
- individuals without smartphones, who do not generate network data,
- individuals carrying multiple smartphones, creating additional network activity independent of unique mobility patterns,
- individuals that are not captured by the RADAR.

Despite these variations, we observe that both datasets generally capture similar trends: when one increases, the other tends to increase, and when one decreases, the other also tends to decrease. However, if

Table 2

Results: Predict handovers using mobility data.

Model	HS Inter-gNB		HS Intra-gNB		Overall	
	AVG R ² -Score	AVG RMSE	AVG R ² -Score	AVG RMSE	AVG R ² -Score	AVG RMSE
LR	0,8416 ± 0,0303	0,0817 ± 0,0163	0,6484 ± 0,2070	0,0860 ± 0,0134	0,7450 ± 0,1001	0,0847 ± 0,0081
Ridge	0,8420 ± 0,0304	0,0816 ± 0,0162	0,6491 ± 0,2059	0,0859 ± 0,0132	0,7455 ± 0,0996	0,0846 ± 0,0081
ElasticNet	0,6690 ± 0,0583	0,1172 ± 0,0184	0,4555 ± 0,2372	0,1106 ± 0,0179	0,5622 ± 0,1226	0,1148 ± 0,0113
DT	0,8630 ± 0,0318	0,0756 ± 0,0151	0,6812 ± 0,1811	0,0824 ± 0,0135	0,7721 ± 0,0894	0,0799 ± 0,0077
RF	0,8954 ± 0,0290	0,0657 ± 0,0132	0,7369 ± 0,1476	0,0748 ± 0,0116	0,8162 ± 0,0737	0,0710 ± 0,0074
XGBoost	0,8964 ± 0,0306	0,0654 ± 0,0138	0,7388 ± 0,1412	0,0750 ± 0,0124	0,8176 ± 0,0716	0,0710 ± 0,0083
LightGBM	0,8968 ± 0,0303	0,0646 ± 0,0138	0,7404 ± 0,1405	0,0745 ± 0,0124	0,8186 ± 0,0708	0,0709 ± 0,0083

**Fig. 4.** 5G intra-gNodeB handovers.

RADARs are deployed in relatively calm areas within a big city, they may not accurately represent overall mobility patterns.

5. From mobility to network and vice-versa

This section presents the results of exploring the relationship between mobility and the network usage. This relation will allow to design models and predict the behavior of one considering the pattern and variations of the other. Namely, we begin by using the RADAR data to predict the handover data (Section 5.1). Then, we use the RADAR data to predict the network traffic load metric (Section 5.2). Finally, we use the network traffic load metric to predict the RADAR data (Section 5.3). Note that we always compare data from both datasets in the same time interval.

In this section, we use the network's most correlated features to predict mobility and vice-versa. More details are given in each subsection.

5.1. Use RADAR data to predict intra and inter-gNB handovers

The handovers are predicted based on the data from the RADARs. We select the RADAR data for each base station with a correlation larger than 0.7 with the fields we want to predict (the handovers intra and inter-gNB). For instance, for station B, we select every RADAR data except RADAR V lane 2 and RADAR W lane 1, given that the correlation values are lower than 0.7 with the handover features. Note that this station is one of the stations that present a higher number of correlations.

Table 2 contains the results of the performance achieved for the handovers inter and intra-gNB and the overall performance. We measure each station's individual performance, presenting the average R²-Score and the average RMSE obtained.

We observe that the performance of predicting the inter-gNB handovers is superior to predicting the intra-gNB handovers. This is a consequence of the characteristics of our data since the inter-gNB handovers are more related to mobility than the intra-gNB handovers.

Table 3

Predict network traffic load using mobility data.

Model	Network traffic load	
	AVG R ² -Score	AVG RMSE
LR	0,8125 ± 0,0495	0,0797 ± 0,0180
Ridge	0,8136 ± 0,0753	0,0835 ± 0,0282
ElasticNet	0,6139 ± 0,0543	0,1143 ± 0,0192
DT	0,8561 ± 0,0486	0,0728 ± 0,0191
RF	0,8761 ± 0,0373	0,0642 ± 0,0131
XGBoost	0,8757 ± 0,0371	0,0642 ± 0,0131
LightGBM	0,8763 ± 0,0383	0,0641 ± 0,0131

For instance, when a user moves from one location to another, they may need to connect to a different cell. This results in inter-gNB handovers. Although the performance of predicting the intra-gNB handovers is lower, we can still achieve an R²-Score of 0.7404 ± 0.1405 using LightGBM, which is good. Regardless, the overall performance is good, and the best model is the LightGBM for all three cases. Nevertheless, the performance achieved using RF and XGBoost is similar to that achieved using LightGBM. The worst method is ElasticNet for all three cases.

5.2. Use RADAR data to predict network traffic load

Similarly to the previous Section 5.1, we evaluate the performance of predicting the network traffic load using the mobility data. Once again, we select the most correlated RADAR features using the same threshold. Table 3 contains the results of predicting the network traffic load using the mobility data.

Considering this case study, the prediction corresponds to the expected values, with an average performance of the R²-Score of 0,8763 ± 0,0383. For this case study, the average R²-Score is slightly lower when compared with the prediction of the inter-gNB handovers, which obtained an R²-Score of 0,8968 ± 0,0303. The LightGBM is the most effective among the models used, while ElasticNet achieved the worst performance.

5.3. Use network traffic load data to predict RADAR data

We decided to invert the prediction direction and predict mobility using network usage data. We follow a similar methodology to the one described in Section 5.1. We select the most correlated network traffic load metrics from the different base stations for a specific RADAR, with a correlation higher than 0.7. Then, we predict the data from the selected RADAR using the network metrics. We repeat the experience for all RADARs.

We compute the average R²-Score and the average RMSE to evaluate our methodology. As we can observe in Table 4, we achieve a worse performance than the inverse scenario in Table 3. The prediction of mobility data using network data, using the best model, achieves an R²-Score of 0,7615 ± 0,1769. In comparison, the prediction of network data using mobility, using the best model, achieves an R²-Score of 0,8763 ± 0,0383. Similarly to the previous results, the best method is the LightGBM and the worst is the ElasticNet.

Table 4

Predict mobility data using network traffic load.

Model	Mobility data	
	AVG R ² -Score	AVG RMSE
LR	0,6721 ± 0,1530	0,1189 ± 0,0196
Ridge	0,6730 ± 0,1522	0,1187 ± 0,0194
ElasticNet	0,5034 ± 0,1701	0,1499 ± 0,0140
DT	0,7022 ± 0,1327	0,1142 ± 0,0177
RF	0,7450 ± 0,1279	0,1044 ± 0,0198
XGBoost	0,7436 ± 0,1314	0,1044 ± 0,0205
LightGBM	0,7615 ± 0,1769	0,0873 ± 0,0133

5.4. Discussion

We can observe some patterns by looking at the three scenarios presented above. In terms of performance, from the worst to the best method, usually the worst method is ElasticNet, followed by Linear and Ridge regression, and then we have the DT; the best methods are RF, XGBoost, and LightGBM, having these three methods a similar performance. Nevertheless, the best overall method is LightGBM.

LightGBM uses gradient boosting to combine several models, usually decision trees. Each new tree learns from the errors of the previous ones. Additionally, LightGBM has several hyperparameters available, including different optimization algorithms. These characteristics make LightGBM more robust and able to capture more complex patterns.

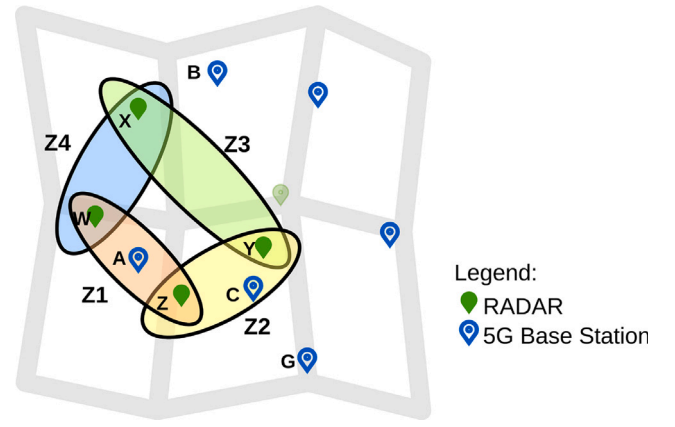
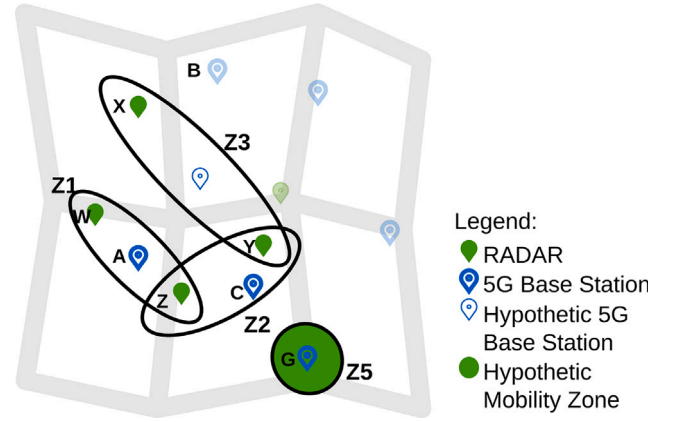
Several applications can benefit from exploring the relationship between the network usage and mobility. For instance, adapting the network infrastructure to local mobility requirements improves the citizens' quality of life. Furthermore, we can use the network information to help guess the missing data or help with forecast tasks in mobility and vice-versa. This approach can enhance the accuracy and effectiveness of various systems that rely on network usage and mobility data. Motivated by the findings of this section, we decided to investigate if it would be possible to extrapolate the relationships found and infer, for instance, the data of new base stations where there is only mobility data. The same can be applied to the opposite case. This study is presented in Section 6.

6. Prediction of 5G network usage and mobility

As we observed in Section 5, we can use the data from one dataset, network or mobility, to predict the data from the other one. This will allow to infer mobility where we do not have mobility stations through the data from 5G communication network usage, and infer the 5G network usage data, through the mobility data, where we do not have 5G network stations.

However, to achieve this, we need to follow a different approach. We decided to group the RADARs and network stations in zones based on their geographical position, the city's morphology, and our knowledge of the datasets. We start by aggregating zones with 2 RADARs. We had to exclude one of the RADARs, given that the road where the RADAR was installed was under construction for some time, meaning that the observed patterns changed. We obtain four possible zones (from Z1 to Z4), presented in Fig. 5. In addition, we create a mobility metric to reduce the information obtained from the RADARs from each zone (both lanes), using PCA for its computation. We observe that, for zones 1, 2, and 3, PCA can explain most of the variance using only one component. The same does not happen for zone 4. The city morphology might explain this. Therefore, we only keep zones 1 to 3. After applying the PCA, we normalize the data between 0 and 1 to create a mobility index. Furthermore, two of the zones of interest intersect with two network stations, one in each zone (zones 1 and 2).

We select three from the four created zones, and only two have data from both datasets. We decided to use zone 3 to infer the data of a hypothetical network base station. Since we aim to perform this study

**Fig. 5.** Schematics of the zones created.**Fig. 6.** Schematics of the aggregations by location.**Table 5**
Zones.

Zone	RADARs	Network station
1	W, Z	A
2	Z, Y	C
3	Y, X	?
5	?	G

in both directions, we selected a network base station from which we would try to infer mobility. Fig. 6 contains a schematic of the zones considered for this part of the study. Table 5 presents some information regarding those zones. From zone 1, we have data from RADARs W and Z and base station A. From zone 2, we have data from RADARs Z and Y and base station C. For zones 3 and 5, we want to infer the data from the network if there would be a network located in that area, and from the RADARs if there would be a RADARs located in that area, respectively.

Sections 6.1 and 6.2 present the two approaches used to infer the network and the mobility. In the first one, we perform a separate train. In the second one, we perform a joint train. More details are given in each section.

6.1. Separated training

We start by trying to infer the network usage data by performing a separate training. We use zone 1 to train the model and evaluate the performance using zone 2. Then, we used zone 2 to train the model and evaluate the performance using zone 1. To select the best model, we compute the average of the performance of each model. Table 6

Table 6

Results: Infer network traffic load from mobility data (LR, Ridge and ElasticNet are deterministic).

Model	Network traffic load					
	Train with Zone 1		Train with Zone 2		Overall	
	R ² -Score	RMSE	R ² -Score	RMSE	AVG R ² -Score	AVG RMSE
LR	0,7574	0,0904	0,6339	0,0924	0,6957 ± 0,0873	0,0914 ± 0,0014
Ridge	0,7577	0,0903	0,6339	0,0924	0,6958 ± 0,0875	0,0914 ± 0,0015
ElasticNet	0,3102	0,1524	0,3869	0,1196	0,3486 ± 0,0542	0,1360 ± 0,0231
DT	0,6452 ± 0,0076	0,1093 ± 0,0011	0,5658 ± 0,0132	0,1007 ± 0,0015	0,5999 ± 0,0482	0,1058 ± 0,0072
RF	0,6797 ± 0,0024	0,1038 ± 0,0003	0,6188 ± 0,0030	0,0943 ± 0,0003	0,6459 ± 0,0477	0,0995 ± 0,0061
XGBoost	0,7345 ± 0,0068	0,0945 ± 0,0012	0,6349 ± 0,0082	0,0923 ± 0,0010	0,6803 ± 0,0767	0,0940 ± 0,0007
LightGBM	0,7353 ± 0,0058	0,0944 ± 0,0010	0,7529 ± 0,0132	0,0759 ± 0,0020	0,6792 ± 0,0793	0,0941 ± 0,0004

Table 7

Results: Infer network traffic load from mobility data, use cases 4 and 5 (LR, Ridge and ElasticNet are deterministic).

Model	Infer network traffic load		Infer mobility	
	R ² -Score	RMSE	R ² -Score	RMSE
LR	0,8910	0,0595	0,8218	0,0978
Ridge	0,8909	0,0595	0,8311	0,0952
ElasticNet	0,6653	0,1042	0,4574	0,1707
DT	0,9014 ± 0,0055	0,0565 ± 0,0016	0,7814 ± 0,0185	0,1083 ± 0,0047
RF	0,9323 ± 0,0008	0,0469 ± 0,0003	0,8866 ± 0,0018	0,0781 ± 0,0006
XGBoost	0,9348 ± 0,0021	0,0460 ± 0,0007	0,8721 ± 0,0031	0,0829 ± 0,0010
LightGBM	0,9388 ± 0,0007	0,0445 ± 0,0002	0,8823 ± 0,0025	0,0795 ± 0,0008

contains the obtained results.

As we can observe, when we train with zone one, the best model is the Ridge regression with an R^2 -Score of 0,7577. Nevertheless, Linear regression achieves a very similar performance, and the best models of LightGBM and XGBoost also achieve a similar performance. When we train with zone 2, LightGBM is, without question, the best model with an R^2 -Score of $0,7529 \pm 0,0132$. However, when we observe the overall performance, LightGBM achieves a lower performance than expected, with an R^2 -Score of $0,6792 \pm 0,0793$. This happens because the training is performed separately; the best parameters that lead to a performance of $0,7353 \pm 0,0058$ when training with zone 1 differ from those that lead to a performance of $0,7529 \pm 0,0132$ when training with zone 2. This approach could have worked if we had more zones with data from both datasets to train our models. Given the results we obtained, we decided to follow a different path.

6.2. Joint training

We will start by explaining the process of inferring the network data. The joint training requires a different approach to organizing the data. We decided to include both zones during the train and the test. Therefore, we take 30% of the data from each zone for the test phase. The first feature is the mobility index, and we add a new feature for the zone. Since, for this case, we will have three different zones, we transform this feature using one hot encoding. After this, we try to predict the network load metric. The test results using this approach are considered good. Nevertheless, we are still trying to improve them by developing and proposing a global mobility metric to use as a new feature. For the global mobility metric, we apply the PCA to the RADARs W, X, Y, and Z.

We observe an increase in the performance presented in Table 7. As we can observe, LightGBM obtains an R^2 -Score of $0,9388 \pm 0,0007$, and it is the best model. Nevertheless, we cannot measure the performance of inferring the network load values.

We follow a similar approach to infer the mobility from the network load data. We also create a global network metric. For this case, we perform the average of the PCA of stations A, B, C, D, and G (stations nearby). As we can observe in Table 7, the random forest achieves an R^2 -Score of $0,8866 \pm 0,0018$. Nevertheless, LightGBM also achieves a very similar performance, achieving an R^2 -Score of $0,8823 \pm 0,0025$.

Although we achieved good results, we cannot measure the accuracy of the hypothetical 5G network base station and mobility metric. Nevertheless, we can visualize the seasonality and other expected patterns by observing the generated values.

6.3. Discussion

The previous sections have shown that we can predict data from one dataset using data from another, enabling us to infer both the mobility and 5G network usage. We grouped RADARs and 5G network stations into zones and created a mobility metric, a global mobility metric, and a global 5G network metric.

In the separate training, using only information from two zones with data from both datasets, we conclude that this approach does not follow the initial requirements, since selecting the best model is biased by the available data. However, we believe that this would be the best approach with a larger dataset, with heterogeneous data, allowing the model to learn from a broad source of information and on a large scale. In the joint training, by implementing a leave one out strategy (in which we select the best model with data from both zones and remove data from each zone for the test phase), we are able to increase the model's performance. However, we cannot measure the performance of inferring the 5G network load values. Nevertheless, we visualize the expected patterns from the generated values, and we show the benefits of studying and relating mobility and 5G network data.

With a large set of 5G stations and mobility-related stations, we would be able to improve the performance and accuracy of the inference of both mobility and 5G network data, enabling the support of an interpolation approach on the analysis of the traffic of both vehicles and network when their probing and analysis cannot take place.

7. Conclusions and future work

The information on both urban mobility and 5G networks and their correlation can benefit both domains. We can identify ways to improve urban spaces and 5G networks by exploring this symbiotic relationship. This study analyzes data from the number of vehicles detected from radars in different places in the city of Aveiro and data from 5G networks on the same interval, with the same periodicity, and with stations near each other. Furthermore, this study examines the

correlation between mobility data and 5G networks. We observed the existence of strong correlations between these two domains.

We then researched an approach to perform the prediction of the 5G network usage with mobility data and vice-versa. We used several machine learning approaches, and we demonstrated that the mobility data can be used to predict the 5G network load and vice versa. To achieve this, we aggregated zones with mobility and network load data and inferred in zones with only one of these components. Based on our observations, we conclude that LightGBM produces good results in most cases, outperforming the other methods. With this work, we created a complete framework for the dynamic symbiosis between urban mobility and 5G networks.

In future work, the relationship between mobility and the 5G network will be further investigated, and additional locations will be analyzed using data from both areas. The dynamic resource and slice allocation provided by NWDAF will serve as valuable information sources that can be integrated into mobility traffic management strategies. Recognizing that multiple factors influence mobility, the impacts of weather, cultural events, and sports activities will also be examined to understand dynamic patterns within the city.

CRediT authorship contribution statement

Ana Almeida: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Pedro Rito:** Writing – review & editing, Validation, Supervision. **Susana Brás:** Writing – review & editing, Validation, Supervision. **Filipe Cabral Pinto:** Writing – review & editing, Validation, Supervision. **Susana Sargento:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

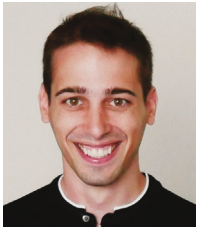
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